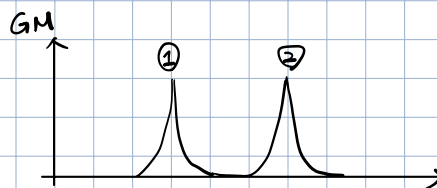
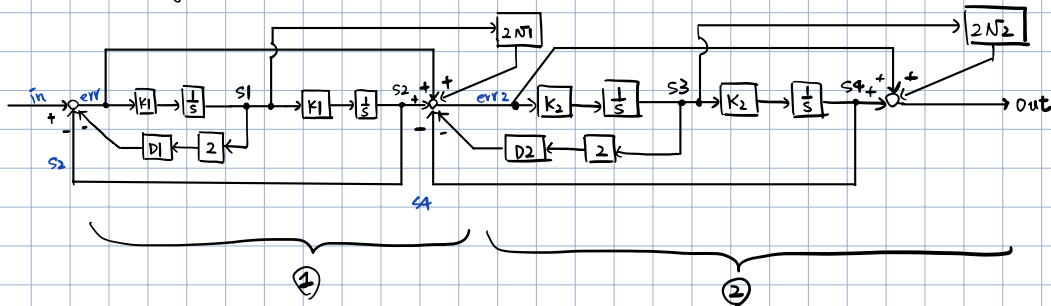


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Block diagram:



如反向的 Notch
只要 $\frac{N_i}{D_i} > 1$

Cogging 為什麼導致 ripple?
且為何用反向 Notch
能夠補償呢?

2. Transfer function

$$G_{\text{ripple}}(s) = \frac{s^2 + 2N_1\omega_n s + \omega_n^2}{s^2 + 2D_1\omega_n s + \omega_n^2}$$

① 與 VSF 相同

② 因 ripple freq 可能有主頻和倍頻,
故串聯了兩個 $G_{\text{ripple}}(s)$

$$\begin{cases} k_1 = 2\pi f_r \\ k_2 = 2 \times k_1 = 2\pi \cdot (2 \cdot f_r) \end{cases}$$

f_r = 低速移動頻率

• 旋轉 $f_r = \text{rpm} \times \text{pole num} \times \frac{1}{120}$

• 線性 $f_r = \frac{\text{mm/s}}{\text{極對距}}$

轉為 Block diagram

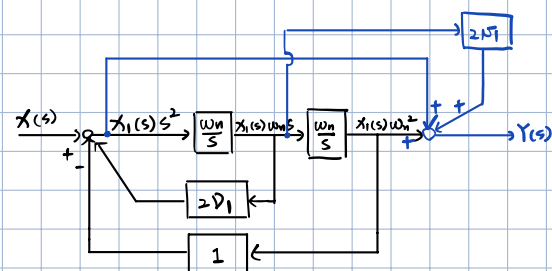
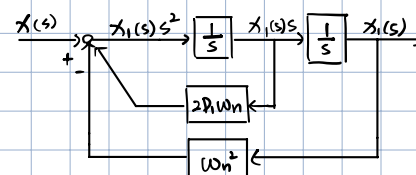
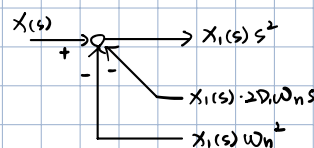
$$X(s) \rightarrow \frac{s^2 + 2N_1\omega_n s + \omega_n^2}{s^2 + 2D_1\omega_n s + \omega_n^2} \rightarrow Y(s)$$

$$X(s) \rightarrow \frac{1}{s^2 + 2D_1\omega_n s + \omega_n^2} \xrightarrow{X_1(s)} \frac{1}{s^2 + 2N_1\omega_n s + \omega_n^2} \rightarrow Y(s)$$

$$X(s) = X_1(s)s^2 + X_1(s) \cdot 2D_1\omega_n s + X_1(s)\omega_n^2$$

$$Y(s) = X_1(s)s^2 + X_1(s)2N_1\omega_n s + X_1(s)\omega_n^2$$

$$X(s) - X_1(s) \cdot 2D_1\omega_n s - X_1(s)\omega_n^2 = X_1(s)s^2$$



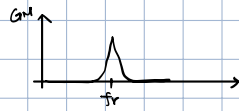
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2. Bode plot

① 反向的 Notch



② D_1, N_1 如何設計，
才能產生反向的 Notch 呢？

$$S = j\omega$$

$$\begin{aligned} \text{Gripple}(s) &= \frac{-\omega^2 + 2jN_1\omega_n \cdot j\omega + \omega_n^2}{-\omega^2 + 2D_1\omega_n \cdot j\omega + \omega_n^2} \\ &= \frac{(\omega_n^2 - \omega^2) + j2N_1\omega_n\omega}{(\omega_n^2 - \omega^2) + j2D_1\omega_n\omega} \end{aligned}$$

$\omega_n = \omega$ 時，

$$= \frac{0 + j2N_1\omega^2}{0 + j2D_1\omega^2}$$

$$\Rightarrow \frac{\sqrt{0^2 + (2N_1\omega^2)^2}}{\sqrt{0^2 + (2D_1\omega^2)^2}}$$

$$\Rightarrow \frac{N_1\omega^2}{D_1\omega^2}$$

$$\Rightarrow \frac{N_1}{D_1} \Rightarrow 20 \log\left(\frac{N_1}{D_1}\right)$$

★ $\frac{N_1}{D_1}$ $\begin{cases} > 1, \log \text{後為正} \Rightarrow \text{ripple 補償} \\ < 1, \quad \quad \quad \text{負} \Rightarrow \text{Notch filter} \\ = 1, \log \text{後為} 0, \text{ i.e. } \text{Out} = \text{in} \end{cases}$

可以先定義 最大補償 的 dB 值

ex. $+1 \text{ dB} \sim +20 \text{ dB}$

若 $N_1 = 1$ (fix)

$$D_1 = \begin{cases} 0.1 \Rightarrow \frac{1}{0.1} \Rightarrow 20 \text{ dB} \\ \vdots \\ 0.5 \Rightarrow \frac{1}{0.5} \Rightarrow 6 \text{ dB} \\ \vdots \\ 0.9 \Rightarrow \frac{1}{0.9} \Rightarrow 0.9 \text{ dB} \end{cases}$$

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3. Code :

```
3. 速度误差补偿
21. Code
float32 err1, K1, err2, K2;
{ K1 = -disf32(2 * N * fr, sample);
  K2 = K1 * 0.5
err1 = in - S2 - (S1 * D1 * 2)
S1 += err1 * K1
S2 += S1 * K1
err2 = highpass (S2 + err1) - S4 + (S1 * N1 - S3 * D2) * 2
S3 += err2 * K2
S4 += S3 * K2
out = S4 + err2 + (S3 * N2 * 2)
```

(S2 + err1 + S1 * 2 * N1)
做第1个 Gripple(s) 输出
并做为第2个 Gripple(s) 的输入